

THE IRRATIONALITY MEASURE OF THE LOGARITHM OF A RATIONAL NUMBER IS 2

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ABSTRACT. For every positive rational $\alpha \neq 1$ the irrationality measure of $\log \alpha$ equals 2: for every $\nu > 2$, $|\log \alpha - p/q| \geq q^{-\nu}$ for all integers p, q with q sufficiently large. In particular $\mu(\log 2) = 2$, where the previous record was 3.5746. The proof is the interpolation-determinant argument that gives $\mu(\pi) = 2$, with the K interpolation centres moved from $(1, 2\pi i j)$ to $(\alpha^j, j \log \alpha)$ on the exponential curve $z \mapsto (e^z, z)$. The single new ingredient is a version of the separated-weight interpolation theorem with centres in distinct fibres of the exponential coordinate; the arithmetic lower bound pays the height of α^j in an error term that the existing parameter choice already drives to zero. For real algebraic α of degree D the same argument gives $\mu(\log \alpha) \leq 2D$, and we explain why ratios of logarithms are not reached. The proof is formalized in Lean 4 over Mathlib, by modifying the formalization of the π theorem module by module.

1. INTRODUCTION

For a real number ω the irrationality measure $\mu(\omega)$ is the supremum of the ν for which $|\omega - p/q| < q^{-\nu}$ has infinitely many rational solutions. Dirichlet gives $\mu \geq 2$ for irrational ω , and $\mu = 2$ for almost every real number. For logarithms of rational numbers the known bounds come from Padé-type constructions: $\mu(\log 2) \leq 3.5746$ (Marcovecchio [3], improving Rukhadze [6] and Hata [2]), and effective bounds of Baker type $\mu(\log \alpha) < \infty$ for every algebraic $\alpha \neq 0, 1$ [1].

The paper [4] proves $\mu(\pi) = 2$ by a Roth-type argument on the exponential curve: an interpolation determinant whose rows are jets at K centres $(1, j\omega)$, $\omega = 2\pi i$, of polynomials restricted to the curve $z \mapsto (e^z, z + u)$, whose columns are weighted monomials, whose arithmetic size is bounded below because the centres are Gaussian integer points after clearing the denominators q_i of the approximations, and whose analytic size is bounded above because rows at different centres test the same entire functions and collide. The theorem is machine-verified in Lean 4 over Mathlib, and so is Theorem A below (Section 6). We cite [4] by its section, equation and result numbers throughout.

Theorem A. *Let α be a positive rational number, $\alpha \neq 1$, and $\omega = \log \alpha$. For every real $\nu > 2$ there is $Q(\nu, \alpha)$ such that*

$$\left| \omega - \frac{p}{q} \right| \geq q^{-\nu} \quad \text{for all integers } p, q \text{ with } q \geq Q(\nu, \alpha).$$

Hence $\mu(\log \alpha) = 2$; in particular $\mu(\log 2) = 2$.

The argument of [4] uses one property of π : $e^{2\pi i j} = 1$, which places the centre $(1, j\omega)$ on the curve at $z = j\omega$ (Lemma 3.2 there). For $\omega = \log \alpha$ the analogous fact is $e^{j\omega} = \alpha^j$, which places the centre $(\alpha^j, j\omega)$ on the curve. The centres therefore lie in distinct fibres of the exponential coordinate Y , and the interpolation theorem of [4], stated for centres with $Y = 1$, must be restated for centres (y_j, c_j) with distinct y_j . Section 2 does this; the proof is that of [4], Section 2, with three local changes which we display. Section 3 carries the Y -height of α^j through the determinant estimates: it costs $(K - 1) \log b/w_0$ in the arithmetic lower bound ($\alpha = a/b$) and replaces the radius $100K$ by $100K(1 + |\log \alpha|)$ in the collision estimate, both of the form $O_\alpha(K/w_0)$, which the parameter choice of [4], Section 4, sends to zero. For

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$\alpha = 2$ the arithmetic lower bound is literally Lemma 3.1 of [4]. Section 5 records what the method gives for algebraic α and why it gives nothing for $\log 3/\log 2$.

Notation. Coordinates $Y, X_1, \dots, X_m$; a positive rational weight vector $W = (w_0, \dots, w_m)$ gives the W -degree $w_0 h + \sum w_i \alpha_i$ of $Y^h X^\alpha$, and $P_W(H)$ is the space of polynomials of W -degree at most H . For $V = (v_0, \dots, v_m)$, $J_V(H)$ is the quotient of $\mathbb{C}[[t, u_1, \dots, u_m]]$ by the series all of whose monomials have V -weight at least H . We write $w\beta = \sum_i w_i \beta_i$, $w_* = \min_i w_i$, $\Lambda = 4 \log 2$, and all logarithms are natural.

2. INTERPOLATION WITH CENTRES IN DISTINCT FIBRES

Theorem 2.1 (Separated-weight interpolation, general fibres). *Fix integers $m, K \geq 1$ and positive rationals w_0, v_0, θ with $0 < \theta < 1$,*

$$(1) \quad K\theta^m < 1, \quad K \frac{w_0}{v_0} \theta^m < 1.$$

There are successive lower thresholds for positive rationals $w_1, \dots, w_m$, each depending only on these data and the previously chosen weights, with the following property. Put $W = (w_0, \dots, w_m)$, $V = (v_0, w_1/\theta, \dots, w_m/\theta)$. For any points

$$a_j = (y_j, c_{j1}, \dots, c_{jm}) \in \mathbb{C}^\times \times \mathbb{C}^m, \quad 0 \leq j < K,$$

such that whenever $y_j = y_{j'}$ with $j \neq j'$ the coordinates $c_{ji}, c_{j'i}$ differ for every i , the map

$$(2) \quad P_W(H) \longrightarrow \bigoplus_{j=0}^{K-1} J_V(H), \quad P \longmapsto \left(P(y_j(1+t), (c_{ji} + u_i + \log(1+t))_{i=1}^m) \right)_{j=0}^{K-1},$$

is surjective for all sufficiently large, sufficiently divisible H . The divisibility can be chosen using only the weights; the remaining threshold for H may depend on the points.

For $y_j = 1$ this is Theorem 2.1 of [4]. When the y_j are pairwise distinct the condition on the c_{ji} is empty.

Proof. We follow [4], Section 2, and indicate the changes. The weights are chosen as in (2.7)–(2.9) there: $\sigma > 0$ small with $(1 + 3\sigma)^{m+1} K(w_0/v_0)\theta^m < 1$, $(1 + 3\sigma)^m K\theta^m < 1$, $(1 + \sigma)\theta < 1$; $v_i = w_i/\theta$; $L_0 = m + 2$; C_* as in (2.8); and $w_1, \dots, w_m$ successively so large that $\prod_{a \in A} w_a > C_* \prod_{b \in B} v_b$ for all index sets A, B of equal size whose largest differing index lies in A . This involves no point.

Let C be an irreducible curve with generic point in the affine chart, with smooth projective model, W -degree

$$\deg_W C = \sum_P \max\left(0, -\frac{\text{ord}_P Y}{w_0}, -\frac{\text{ord}_P X_1}{w_1}, \dots, -\frac{\text{ord}_P X_m}{w_m}\right)$$

as in (2.10), and, at a branch P through a_j ,

$$(3) \quad h_P = \min\left(\frac{\text{ord}_P(Y - y_j)}{v_0}, \frac{\text{ord}_P(X_1 - c_{j1} - \log(Y/y_j))}{v_1}, \dots, \frac{\text{ord}_P(X_m - c_{jm} - \log(Y/y_j))}{v_m}\right),$$

where $\log(Y/y_j)$ is its formal expansion at $Y = y_j$, well defined along the branch since $Y - y_j$ has positive order there. This replaces (2.11), where $y_j = 1$. In the local coordinate $t = Y/y_j - 1$ at a_j the jet (2) is $P(y_j(1+t), c_j + u + \log(1+t))$, so h_P measures, in V -weight, the contact of the branch with the jet. The frame $D_0 = Y\partial_Y + \sum_i \partial_{X_i}$, $D_i = \partial_{X_i}$ of [4] is unchanged. In the coordinates (t, u) at a_j , where $Y = y_j(1+t)$ and $X_i = c_{ji} + u_i + \log(1+t)$, the chain rule gives $\partial_t = y_j\partial_Y + (1+t)^{-1} \sum_i \partial_{X_i}$, hence $(1+t)\partial_t = D_0$ and $\partial_{u_i} = D_i$: the action of the frame on jets is the same at every centre, which is all that the order bookkeeping of [4] (the display after (2.13), “each monomial has order at least its weight times h_P ”) uses.

Curve inequality. We claim

$$(4) \quad \deg_W C \geq (1 + \sigma) \sum_{P \mapsto a_j} h_P,$$

the sum over all branches through prescribed points. Suppose it fails. The construction of the auxiliary polynomial F_N (of W -degree at most N , V -order at least $(1 + 3\sigma)N$ at every centre) is a count of coefficients against jet conditions; the number of jet conditions at a_j is the number of monomials $t^s u^\beta$ of V -weight below $(1 + 3\sigma)N$, independent of y_j , so the count (2.13) is unchanged. The zero–pole comparison on C (poles of $F_N|_C$ of total degree at most $N \deg_W C$, zeros of order at least $(1 + 3\sigma)N h_P$ at the branches through centres) uses only (3) and the order bookkeeping, and gives $F_N|_C = 0$ together with the vanishing (2.14) of all frame derivatives of cost at most σN on C° . The persistent component $Z \supseteq C^\circ$, the normal-basis inequality (2.15) from Lemma 2.2 there, the weight separation, and the residue argument are statements about Z , F_N and the frame, and involve no centre. They give: Y is constant on Z .

Here is the first change. In [4] one concludes “ $Y = 1$ on C° ” because C meets a centre with $Y = 1$. Now $C^\circ \subseteq Z$ meets some centre a_{j_0} (otherwise there is nothing to prove), so $Y \equiv y_{j_0}$ on C . Consequently every centre that C meets lies in the fibre $Y = y_{j_0}$, and the sum in (4) runs over branches through centres of that fibre only.

The fibre argument. Work in the affine fibre $Y = y_{j_0}$, in which $h_P = \min_i \text{ord}_P(X_i - c_{ji})/v_i$ for the centres a_j of the fibre (the Y -term of (3) is $+\infty$ and $\log(Y/y_j) = 0$ there). This is the second change: in [4] all K centres lie in the fibre, here only those with $y_j = y_{j_0}$, say $K' \leq K$ of them. The auxiliary polynomial $G_N(X_1, \dots, X_m)$ of weighted degree at most N with order at least $(1 + 3\sigma)N$ at these K' centres exists because the ratio of conditions to coefficients tends to $K'(1 + 3\sigma)^m \theta^m \leq K(1 + 3\sigma)^m \theta^m < 1$. The zero–pole comparison, the chain U_r , the persistent component Z' of codimension $1 \leq k \leq m - 1$ (for $m \geq 2$; for $m = 1$ the contradiction is immediate), and the unique normal index set are as in [4]; they give that some X_i is constant on Z' .

Third change: “coordinatewise distinctness shows Z' meets at most one prescribed point” is the hypothesis of Theorem 2.1 restricted to the fibre: centres sharing y_{j_0} have distinct i -th coordinates. When the y_j are pairwise distinct, $K' = 1$, the fibre has a single centre, and this second stage (the polynomial G_N , the chain U_r , the component Z') is not needed at all. In either case C meets only one centre a_j of the fibre, and with a coordinate X_i nonconstant on C ,

$$\deg_W C \geq \theta^{-1} \sum_{P \mapsto a_j} h_P > (1 + \sigma) \sum_{P \mapsto a_j} h_P$$

by $(1 + \sigma)\theta < 1$, contradicting the failure of (4).

From the curve inequality to jets. Section 2.3 of [4] works on the weighted projective closure with the weighted point ideals $I_j = (t^{R/v_0}, (u_i')^{R/v_i})$ at the centres, the blow-up along them, the nefness of $A - (1 + \sigma)E$ (from (4) applied to every curve, including curves missing the chart, which are handled there without reference to the centres), ampleness of $A - E$, and Serre vanishing giving $H^1 = 0$ and surjectivity onto the truncated jets; the final step undoes the truncation by the invertible filtered substitution. The only point-dependent object is the local coordinate $t = Y - 1$ at the centre, which becomes $t = Y/y_j - 1 = (Y - y_j)/y_j$, a regular parameter at a_j since $y_j \neq 0$; (2.18)–(2.23) there are verbatim. The thresholds for $w_1, \dots, w_m$ come from (2.7) and (2.9) and involve no point; the threshold for H (through the choice of N and the vanishing) may depend on the points. $\square$

3. THE DETERMINANT

Let $\alpha = a/b$ with coprime positive integers $a \neq b$, $\omega = \log \alpha \neq 0$. Let $\nu > 2$ and suppose integers $p_i, q_i \geq 2$ satisfy

$$(5) \quad \left| \omega - \frac{p_i}{q_i} \right| \leq q_i^{-\nu}, \quad w_i = \lceil \log q_i \rceil, \quad r_i = \frac{p_i}{q_i} \quad (1 \leq i \leq m),$$

with $m, K, w_0, v_0, \theta, w_1, \dots, w_m$ admissible for Theorem 2.1. Since $\omega \neq 0$, $p_i \neq 0$ once q_i is large. Fix $F_0 > 2/\theta$, $T_i = F_0 w_i / v_0$, and the truncated logarithms $G_i(t) = \sum_{1 \leq k < T_i} (-1)^{k+1} t^k / k$ of (3.2) in [4]; replacing $\log(1+t)$ by G_i is an invertible filtration-preserving substitution, so Theorem 2.1 remains surjective with the G_i (Section 3 of [4]).

Columns are the monomials $Y^h X^\alpha$ with $w_0 h + w\alpha \leq H$; rows are the triples $\rho = (j, s, \beta)$ with $0 \leq j < K$, $v_0 s + w\beta/\theta < H$; the entry is the coefficient of $t^s u^\beta$ in

$$(6) \quad P(\alpha^j(1+t), jr_1 + G_1(t) + u_1, \dots, jr_m + G_m(t) + u_m).$$

By Theorem 2.1 with centres $(\alpha^j, jr_1, \dots, jr_m)$, whose Y -coordinates α^j are pairwise distinct because $\alpha > 0, \alpha \neq 1$, for all large divisible H there is a nonzero square minor Δ_H using every row. Its size $M = M_H$ satisfies (3.5) of [4], $M \sim K\theta^m H^{m+1} / ((m+1)! v_0 \prod_i w_i)$, and $\bar{b} = \bar{b}_H = (MH)^{-1} \sum_\rho w\beta \in [0, \theta]$.

Lemma 3.1 (Arithmetic lower bound). *With $E_{\text{ar}} = \Lambda F_0 m / v_0 + \Lambda \sum_i 1/w_i + \theta/w_*$ as in (3.6) of [4] and*

$$E'_{\text{ar}} = E_{\text{ar}} + \frac{(K-1) \log b}{w_0},$$

every minor chosen as above satisfies $\log |\Delta_H| / (MH) \geq -(1 - \bar{b}) - E'_{\text{ar}}$.

Proof. For $P = Y^h X^\alpha$ the entry of row (j, s, β) equals

$$\alpha^{jh} \prod_i \binom{\alpha_i}{\beta_i} [t^s] (1+t)^h \prod_i (jr_i + G_i(t))^{\alpha_i - \beta_i}$$

when $\alpha \geq \beta$ and 0 otherwise. Scale column $Y^h X^\alpha$ by $\prod_i q_i^{\alpha_i}$, row (j, s, β) by $\prod_i q_i^{-\beta_i}$, and multiply by $D_H = \prod_i \text{lcm}(1, \dots, T_i - 1)^{\lfloor H/w_i \rfloor}$, which clears the denominators of the G_i (proof of Lemma 3.1 in [4]); since $q_i jr_i = jp_i \in \mathbb{Z}$, all factors except α^{jh} are then integers. Put $h_{\max} = \lfloor H/w_0 \rfloor$ and scale row (j, s, β) additionally by $b^{jh_{\max}}$, turning α^{jh} into $a^{jh} b^{j(h_{\max} - h)} \in \mathbb{Z}$. The scaled matrix is over $\mathbb{Z}$ with nonzero determinant, so its determinant has modulus at least 1. The row scalings by $b^{jh_{\max}}$ contribute at most $\sum_\rho j h_{\max} \log b \leq M(K-1)(H/w_0) \log b$ to the logarithm; the remaining scalings are accounted for exactly as in (3.8) of [4]: $\sum_{\text{cols}} \sum_i \alpha_i \log q_i \leq MH$, $\sum_{\text{rows}} \sum_i \beta_i \log q_i \geq MH\bar{b} - MH\theta/w_*$, and $\log D_H \leq MH(\Lambda F_0 m / v_0 + \Lambda \sum_i 1/w_i)$. $\square$

For $\alpha = 2$ one has $b = 1$ and Lemma 3.1 is Lemma 3.1 of [4] with $\mathbb{Z}$ in place of $\mathbb{Z}[i]$.

Lemma 3.2 (Exact row translation). *Put $z = j\omega + \log(1+t)$. Then $e^z = \alpha^j(1+t) = Y$ and $X_i = z + u_i + \varepsilon_i + \tau_i(t)$ with $\varepsilon_i = j(r_i - \omega)$, $\tau_i = G_i - \log(1+t)$. Every row of (6) is a sum of at most $Q_H = (\lfloor H \rfloor + 1)^{2m} (\lfloor H/v_0 \rfloor + 1)$ scalar multiples of the test rows $([t^\ell] f_{a,P}(j\omega + \log(1+t)))_P$, $a \geq \beta$, $wa \leq H$, $0 \leq \ell \leq s$, where*

$$f_{a,P}(z) = [u^a] P(e^z, z + u_1, \dots, z + u_m)$$

is an entire function of z independent of j , and a scalar with index a satisfies $|\xi| \leq \exp\{-\nu w(a - \beta) + HE_{\text{tr}}\}$ with $E_{\text{tr}} = \nu/F_0 + \log 2/v_0 + (\log 4 + \log(2K) + \nu)/w_$ as in (3.12) of [4].*

Proof. The identity $e^{j\omega} = \alpha^j$ gives $e^z = Y$; the rest is the substitution of Lemma 3.2 in [4], and the identity (3.13) and its bounds are verbatim, with $|\varepsilon_i| \leq (K-1)q_i^{-\nu}$, which is smaller than the bound $2Ke^\nu e^{-\nu w_i}$ used there. $\square$

Lemma 3.3 (Collision estimate). *Put $R_\alpha = 100K(1 + |\log \alpha|)$, $c = (\log 2)/4$ and*

$$E'_{\text{hol}} = \frac{R_\alpha}{w_0} + \frac{\log 2}{v_0} + \frac{\log(2R_\alpha)}{w_*}.$$

Choose one translated test row from each row expansion of Lemma 3.2, remove the scalars, and let T be the resulting matrix and n_a the number of rows with transverse index a . Then $|\det T| \leq \exp\{-c \sum_a n_a^2 + MH(E'_{\text{hol}} + \rho_H)\}$ with $\rho_H \rightarrow 0$ independent of the choices.

Proof. On $|z| \leq R_\alpha$ one has $|f_{a,P}(z)| \leq \exp\{H(R_\alpha/w_0 + \log(2R_\alpha)/w_*)\}$ by (3.10) of [4], since $h \leq H/w_0$ and $|\alpha| \leq H/w_*$. On $|t| = 1/2$, $|j\omega + \log(1+t)| \leq (K-1)|\log \alpha| + \log 2 \leq R_\alpha/2$. The expansion of each $f_{a,P}$ in powers of z/R_α , the vanishing of determinant terms with a repeated pair (a, d) , the bound $\sum_\rho d_\rho \geq \sum_a \binom{n_a}{2}$, and the geometric decay are those of Lemma 3.3 in [4] with R replaced by R_α . $\square$

Proposition 3.4 (Comparison). *Let $0 < \eta < 1$, $A \in (0, 1)$, $L = \eta^2 K \theta^m / ((m+1)v_0 A^m)$, $E'_{\text{an}} = E'_{\text{tr}} + E'_{\text{hol}}$, and*

$$g = \nu(A(1-\eta) - \theta) - (1-\theta).$$

If $g > 0$, $E'_{\text{ar}} + E'_{\text{an}} < g$ and $cL > 1 + E'_{\text{ar}} + E'_{\text{an}}$, then for all sufficiently large divisible H the bounds of Lemma 3.1 and Lemmas 3.2–3.3 are incompatible.

Proof. Proposition 3.4 of [4], whose proof uses only the three lemmas, the simplex count $D_A(H) \sim (AH)^m / (m! \prod w_i)$ of rows with $wa \leq AH$, and $\bar{b} \leq \theta$: either at least ηM rows have $wa \leq AH$, and Cauchy–Schwarz gives $\sum_a n_a^2 \geq MH(L - o(1))$, so Lemma 3.3 beats Lemma 3.1 when $cL > 1 + E'_{\text{ar}} + E'_{\text{an}}$; or more than $(1-\eta)M$ rows have $wa > AH$, and the scalars of Lemma 3.2 supply $\exp\{-\nu(A(1-\eta) - \bar{b})MH\}$, which beats Lemma 3.1 when $\nu(A(1-\eta) - \bar{b}) - (1-\bar{b}) \geq g > E'_{\text{ar}} + E'_{\text{an}}$, the minimum over $\bar{b} \in [0, \theta]$ being at $\bar{b} = \theta$ since $\nu > 1$. $\square$

4. PROOF OF THEOREM A

Lemma 4.1 (Parameters; Lemma 4.1 of [4]). *Let $\nu > 2$. There are rationals $0 < \theta < A < B < 1$ and $C > 1$ with*

$$\nu(A - \theta) > 1 - \theta, \quad A^2 < \theta, \quad B < 1/C, \quad B < C\theta.$$

Proof. Take $b \in (1/2, 1 - 1/\nu)$, $\theta = 1 - \delta$, $A = 1 - b\delta$ with δ small: then $\nu(A - \theta) - (1 - \theta) = (\nu(1 - b) - 1)\delta > 0$ and $\theta - A^2 = (2b - 1)\delta - b^2\delta^2 > 0$; since $A < \sqrt{\theta}$ one has $A/\theta < 1/A$, so $C \in (A/\theta, 1/A)$ and then $B \in (A, \min(1/C, C\theta))$ exist. $\square$

Proof of Theorem A. Fix $\nu > 2$ and suppose, for contradiction, that arbitrarily large q admit p with $|\omega - p/q| \leq q^{-\nu}$. Take θ, A, B, C from Lemma 4.1, then $\eta > 0$ so small that $g = \nu(A(1-\eta) - \theta) - (1-\theta) > 0$, put $\varepsilon_0 = \min(g/2, 1/2)$, and fix $F_0 > 2/\theta$ with $\nu/F_0 < \varepsilon_0/3$. For an integer m put

$$K = \lfloor C^m \rfloor, \quad w_0 = B^{-m}, \quad v_0 = 2K\theta^m w_0.$$

Then $K\theta^m < 1$ and $K(w_0/v_0)\theta^m = 1/2$, so (1) holds; $K/w_0 \leq (CB)^m \rightarrow 0$; $v_0 \sim 2(C\theta/B)^m \rightarrow \infty$; and $L = \eta^2(B/A)^m/(2(m+1)) \rightarrow \infty$ as $m \rightarrow \infty$. Choose m so large that

$$(7) \quad \frac{\Lambda F_0 m + 2 \log 2}{v_0} + \frac{100K(1 + |\log \alpha|) + (K-1) \log b}{w_0} < \frac{\varepsilon_0}{3}, \quad cL > 2;$$

this is possible because the second summand is at most $(CB)^m(100(1 + |\log \alpha|) + \log b)$. Fix m, K, w_0, v_0 , hence the Y -coordinates α^j of the centres. Choose approximations p_i/q_i satisfying (5), $w_i = \lfloor \log q_i \rfloor$, successively so large that they meet the thresholds of Theorem 2.1 (which depend only on the fixed data and the earlier w 's) and

$$(8) \quad \Lambda \sum_i \frac{1}{w_i} + \frac{\theta + \log 4 + \log(2K) + \nu + \log(2R_\alpha)}{w_*} < \frac{\varepsilon_0}{3}.$$

Then $E'_{\text{ar}} + E'_{\text{an}} < \varepsilon_0 \leq g/2$ and $cL > 2 > 1 + E'_{\text{ar}} + E'_{\text{an}}$, and Proposition 3.4 gives a contradiction for large H . Hence $|\omega - p/q| > q^{-\nu}$ for all $q \geq Q(\nu, \alpha)$. Taking $\nu = 3$ shows that ω is not rational (a rational $\omega = p/q$ would give infinitely many exact representations p'/q'), so $\mu(\omega) \geq 2$ by Dirichlet, and $\mu(\omega) = 2$. $\square$

The threshold $Q(\nu, \alpha)$ is ineffective, exactly as in [4]: the dimension m and the approximations are chosen against a hypothetical sequence of exceptionally good approximations.

Corollary 4.2. *For every positive rational $\alpha \neq 1$ and every $\nu > 2$, $\|q \log \alpha\| \geq q^{1-\nu}$ for all large q , where $\|\cdot\|$ is the distance to the nearest integer; equivalently $|\alpha^q - e^p| \geq \frac{1}{2}e^p q^{1-\nu}$ for all large q and all p .*

5. ALGEBRAIC α , AND THE LIMITS OF THE METHOD

Proposition 5.1. *Let $\alpha > 0$ be a real algebraic number of degree D , $\alpha \neq 1$. Then $\mu(\log \alpha) \leq 2D$.*

Proof sketch. Run Sections 3–4 with centres (α^j, jr_i) . After the column and row scalings and an extra row scaling by $d^{jh_{\max}}$, where d is a denominator of α , the matrix has entries in the ring of integers of $\mathbb{Q}(\alpha)$, and $\Delta' = (\text{scalings}) \cdot \Delta_H$ is a nonzero algebraic integer, so $|N_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\Delta')| \geq 1$, i.e. $\log |\Delta_H| \geq -DS - \sum_{\sigma \neq \text{id}} \log |\sigma \Delta_H|$ with $S = MH(1 - \bar{b}) + MHE'_{\text{ar}}$. The conjugate $\sigma \Delta_H$ is the determinant of the same matrix with α replaced by $\sigma(\alpha)$; its rows are jets of $P(\lambda^j e^z, z + u + \varepsilon + \tau)$ with $\lambda = \sigma(\alpha)/\alpha$ and $z = j\omega + \log(1 + t)$, so the column $Y^h X^\alpha$ tests $\lambda^{jh} f_{a,P}(z)$: the factor λ^{jh} depends on the row through j and on the column through h , and rows at different centres with the same (a, d) are no longer proportional. Only Hadamard's bound $\log |\sigma \Delta_H| \leq MH \cdot o(1)$ is available for the conjugates. The comparison then needs $\nu(A(1 - \eta) - \theta) - D(1 - \theta) > 0$, and with $\theta = 1 - \delta$, $A = 1 - b\delta$, $b > 1/2$ this is $\nu(1 - b) > D$, giving every $\nu > 2D$. $\square$

If λ were a root of unity of order n , grouping the rows by $j \bmod n$ would restore the collision at the cost $L \rightarrow L/n$; but $\sigma(\alpha) = \zeta \alpha$ for all σ forces $\alpha^n \in \mathbb{Q}$, which is the rational case.

Remark 5.2 (Ratios of logarithms). *The method does not reach $\xi = \log 3 / \log 2$ or any ratio of logarithms, for which the known bounds come from linear independence measures for $1, \log 2, \log 3$ (Rhin [5]; see Zudilin [7] for the state of the art and the method). An approximation p/q to ξ is the statement $3^q / 2^p = e^\varepsilon$ with $|\varepsilon| \leq q^{1-\nu} \log 2$. The method requires, for each approximation, an algebraic centre on the curve $z \mapsto (e^z, z + u)$ with a rational jet whose arithmetic cost matches the weights: X -heights $\log q_i$ against X -degrees of weight $w_i = \log q_i$, and $O(K)$ Y -heights against Y -degrees $h \leq H/w_0$ with w_0 fixed. For $\log 2$ the algebraic object attached to p/q is p/q itself, of height $\log q$, and the error $q^{-\nu}$ is exponentially small in that height. For ξ the algebraic object is $3^q / 2^p$, of height $\asymp q$, against the error $q^{1-\nu}$, polynomially small in the height: placing it in a centre costs $2q \log 3 / w_0$ per unit of H , unbounded, while placing the approximation in the jet (the curve $Y = 2^X$, whose tangent field $Y \partial_Y + (\log 2)^{-1} \partial_X$ is not defined over $\mathbb{Q}$) costs denominators q^s in the derivative direction, whose weight v_0 is fixed. This is Baker's regime, not the Roth regime of the determinant. The number e^ξ is transcendental by Gelfond–Schneider, so no centre of the exponential curve attached to ξ is algebraic.*

Remark 5.3 (Falsification tests). *The argument must fail for $\alpha = 1$: then $\omega = 0$, $p_i = 0$ for every good approximation, and the centres $(1, 0)$ coincide, violating the hypothesis of Theorem 2.1; and Theorem A is false for $\omega = 0$. It must fail for a general transcendental ω , since Liouville numbers exist: for such ω the point $e^{j\omega}$ is not algebraic, and Lemma 3.1 has no integral matrix. For $\omega = \log \alpha$ with α algebraic, Baker's theorem gives $\mu(\omega) < \infty$, consistent with Proposition 5.1, and $\log \alpha$ is never a Liouville number.*

6. FORMAL VERIFICATION

The directory `lean/` accompanying this paper contains a Lean 4 development over Mathlib (Lean v4.34.1, Mathlib at the commit pinned by [4]). The root module `LogIrrationalityMeasure.lean` proves

```
theorem irrationality_measure_log (α : ℚ) (hα : 0 < α) (hα1 : α ≠ 1) :
  (∀ ν : ℝ, 2 < ν → ∃ Q : ℤ, 2 ≤ Q ∧
    ∀ p q : ℤ, Q ≤ q → (q : ℝ) ^ (−ν) ≤ |Real.log α − (p : ℝ) / (q : ℝ)|) ∧
  sSup {ν : ℝ | 0 < ν ∧ Set.Infinite {r : ℚ | 2 ≤ r.den ∧
    0 < |Real.log α − (r : ℝ)| ∧ |Real.log α − (r : ℝ)| < (r.den : ℝ) ^ (−ν)}} = 2
```

```
theorem log_not_liouvilleWith (α : ℚ) (hα : 0 < α) (hα1 : α ≠ 1) {p : ℝ} (hp : 2 < p) :
  ¬ LiouvilleWith p (Real.log α)
```

The first is Theorem A in the vocabulary of the challenge file of [4]: for every real $\nu > 2$ an integer $Q \geq 2$ with $q^{-\nu} \leq |\log \alpha - p/q|$ for all integers p, q with $q \geq Q$, and the supremum of the set of $\nu > 0$ admitting infinitely many rationals r with $0 < |\log \alpha - r| < \text{den}(r)^{-\nu}$ equal to 2. That supremum is `irrationalityExponent` in [4], whose definition is copied unchanged; `LiouvilleWith` is Mathlib's. The same module states the case $\alpha = 2$ as `irrationality_measure_log_two`.

The proof, under `LogIrrationalityMeasure/`, is the formalization of [4] modified module by module: 863 files, 94,643 lines. The geometric side (Theorem 2.1) is the development of [4], Section 2, with the base point $Y = 1$ of the centres replaced by a parameter y and the j -th centre placed at y^j ; the arithmetic side is Lemmas 3.1–3.3, Proposition 3.4 and Section 4 as written, with the row scaling $b^{jh_{\max}}$, the identity $e^{j\omega} = \alpha^j$, the radius $100K(1 + |\omega|)$ and the error term $(K - 1) \log b/w_0$ where the text puts them. Every theorem above depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`, as `AxiomCheck.lean` verifies.

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